

Dave's Daily

AI • Bayesian optimization • US equities

tl;dr

1. OpenAI began watermarking ChatGPT text in the EU. Its own tests show how fragile the mark is: swapping a quarter of the words for synonyms cuts detection from 92% to 17%. DeepMind's new watermark for AI-designed proteins can also be scrubbed.
2. Stocks sit near records with the 10-year Treasury above 5.3%, but the S&P 500 has quietly cheapened, from over 21 times forward earnings in May to under 19. Its earnings yield now roughly equals the Treasury yield.
3. A new paper finds that high-dimensional Bayesian optimization often works better with one shared lengthscale than with one per input. With few data points, the likelihood cannot see most of the per-input lengthscales.

AI

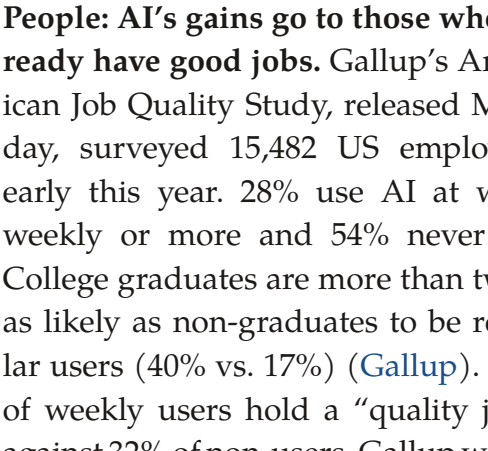
A watermark in the word choices. On Monday OpenAI said it will add an invisible watermark, called textGrain, to ChatGPT and Codex text in the EU over the coming weeks, to comply with the EU AI Act's transparency rules, which took effect on 2 August. API customers anywhere can opt in; it stays off by default, and access to the detector goes first to approved researchers ([OpenAI](#); [TechCrunch](#)). The mark nudges which word the model samples, so it survives copy and paste. A [technical report](#) shows that the detector needs only the text and a secret key. Short passages, math, and light paraphrase weaken it (chart).

Proteins get one too. DeepMind's SynthID Bio, published in [Nature](#), steers amino-acid choices in AI-designed proteins. In wet-lab tests against three targets (VEGF-A, the SARS-CoV-2 spike, and PD-L1), watermarked binders matched unmarked ones on hit rate and binding affinity ([DeepMind](#)). The team concedes the mark is relatively easy to scrub ([Science](#)). Hypothesis: both watermarks label honest output by default; neither stops a motivated evader.

An American open model, at a fraction of the compute. Reflection AI, valued at \$25 billion and backed by Nvidia, unveiled Beam: a mixture-of-experts model with 501 billion parameters, 23 billion of them active per token. Its weights are due this month under the Apache 2.0 license ([Reflection](#); [TechCrunch](#)). Reflection says Beam matches China's GLM-5.2 on reasoning with 3 to 4 times less inference compute. Nobody has verified that claim yet. Also Monday: ChatGPT will start showing display ads next to generated images in the US ([TechCrunch](#)).

How sturdy is ChatGPT's watermark?

Share of watermarked passages detected, %, at a 1% false-positive target



Navy: passage length, psychology text, no edits. Gold: synonym edits to 400-token passages. Two separate tests by OpenAI, not independently replicated. Math text is detected less often. The axis starts at zero. Source: OpenAI, "Our approach to EU text provenance rules", 5 October 2026.

People: AI's gains go to those who already have good jobs. Gallup's American Job Quality Study, released Monday, surveyed 15,482 US employees early this year. 28% use AI at work weekly or more and 54% never do. College graduates are more than twice as likely as non-graduates to be regular users (40% vs. 17%) ([Gallup](#)). 52% of weekly users hold a "quality job," against 32% of non-users. Gallup warns this is correlation: the same groups have both. One cost: 31% of AI users say their employer now asks more of them, rising to 38% among 18- to 24-year-olds. And 52% of all employees want more say over which technology their workplace adopts.

Markets

Why 5.3% yields haven't sunk stocks.

The 10-year Treasury yield topped 5.3% last week and the 30-year reached 5.69%, levels last seen in 2002. Yet the Nasdaq 100 set a record on Friday ([Bloomberg via Moneycontrol](#)). Earnings are doing the work. Bloomberg Intelligence expects third-quarter S&P 500 earnings per share up more than 24%, and tech up more than 65%. Meanwhile the index's forward price-to-earnings ratio fell from over 21 in May to under 19, so prices have lagged profits. At 19 times earnings, the earnings yield is $1/19 \approx 5.3\%$, about the same as the 10-year Treasury. Hypothesis: investors are accepting almost no extra expected return for owning stocks over bonds, a bet that earnings keep growing. Unlike in 2023, Alphabet, Amazon, and Meta now have negative annual free cash flow.

AI fear makes software cheap, and the buyer paid for it. Schneider Electric, which supplies power and cooling gear to data centers, agreed to buy PTC, the product-design software maker, for \$205 a share in cash: \$22.6 billion, a 42.3% premium ([deal release](#)). Jefferies says fears that AI will disrupt software let Schneider buy PTC at a decade-low valuation. Even so, Schneider's shares fell nearly 10%, erasing close to €15 billion, about three-quarters of the price ([Reuters](#)). It will borrow €16 to 17 billion to pay.

Backtests that fight back. Most backtests replay history, so your orders cannot move the market. BMLL, a supplier of order-book data, and Simudyne are training generative "Large Market Models" on full order-book histories (Level 2 and Level 3, down to individual orders). The resulting simulator reacts to the user's orders in real time ([BMLL](#); [FOW](#)). For high-frequency and execution desks, market impact is the hard part to test. Hypothesis: the value depends on whether simulated impact matches live fills, and neither firm has published that evidence.

Key numbers (this edition)

Item	Figure
Watermark detected, 25% of words swapped	17%
S&P 500 forward P/E, May to now	21 → <19
Expected Q3 S&P 500 EPS growth	>24%
Schneider one-day share drop	≈ −10%
US employees using AI weekly	28%

Bayesian optimization

Scaling: one lengthscale instead of thousands.

Standard Bayesian optimization fits a Gaussian process with automatic relevance determination (ARD), one lengthscale per input. Wei-Ting Tang, Madhav Muthyala, and Joel Paulson (Wisconsin–Madison) ask whether limited data can support that ([arXiv:2610.05780](#), 5 Oct).

The mechanism. For radial kernels, the marginal likelihood sees the lengthscales only through weighted pairwise distances between observed points. That is a linear map from D weights to $n(n-1)/2$ distances. With the usual 10 starting points there are 45 pairs, so in 100 dimensions at least 55 directions of lengthscale change leave the likelihood exactly unchanged. Those changes still alter predictions at new points. Even with more data, some directions stay weakly constrained.

The fix. Iso-BO keeps the rest of a modern "vanilla" pipeline (dimension-scaled priors, LogEI acquisition, and RAASP-style starting points) but fits one shared lengthscale.

The evidence. The authors ran 1,000 evaluations on six real problems from 100 to 6,392 dimensions. Iso-BO clearly beat its matched ARD twin on HalfCheetah and HumanoidStandup, with weaker evidence on Rover. ARD leaned ahead on a 388-dimension SVM task, and Humanoid was mixed. It often won even on objectives built to be anisotropic, though ARD won under the strongest anisotropy tested. It also fits hyperparameters faster. There is no regret proof, and the code awaits acceptance. Hypothesis: as with dimension-scaled priors, much of high-dimensional BO's gain comes from asking the surrogate to learn less, not more.

Updated Tuesday 6 October 2026. Same edition ships as phone and letter. Most items are from 5 October; SynthID Bio and the BMLL deal date from 30 September, and Gallup's survey ran January to March. Market notes cover mechanisms, not levels or forecasts. Created with [malvin](#).